

## School of Computer Applications and Technology

Semester End Examination - Jan 2026

### Masters of Computer Applications

Duration : 120 Minutes  
Max Marks : 50

#### Sem I - E1PA101T - Computational Mathematics and Statistics

##### General Instructions

Answer to the specific question asked

Draw neat, labelled diagrams wherever necessary

Approved data hand books are allowed subject to verification by the Invigilator

- 1)  $C = \begin{bmatrix} 1 & 2 \\ 7 & 1 \end{bmatrix}, D = \begin{bmatrix} 11 & 4 \\ 5 & 8 \end{bmatrix}, E = \begin{bmatrix} 1 & 4 \\ 5 & 1 \end{bmatrix}$ , find following value of the matrix  $10C - 3D + 5E$  using the matrix operations multiplications, additions and subtractions on C, D and E. CO1(5)
- 2) Find the standard deviation for the following individual series 3, 6, 8, 10, 12, 14. CO2(5)
- 3) Define conditional probability. The probability of a student passing in science is  $\frac{4}{5}$  and the of the student passing in both science and maths is  $\frac{1}{2}$ . What is the probability of that student passing in maths knowing that he passed in science. CO3(5)
- 4) State Central limit theorem. The lifetimes of electric bulbs produced by a factory have  $\mu=1000$  hours,  $\sigma=100$  hours. A random sample of 36 bulbs is selected. Find the probability that the sample mean lifetime is between 980 and 1020 hours. Given the value from standard normal tables:  $P(-1.2 < Z < 1.2) = 0.884$ . CO4(7)
- 5) Find the equation of line for data x and y CO2(7)  

|   |   |   |   |    |    |
|---|---|---|---|----|----|
| X | 3 | 6 | 9 | 12 | 15 |
| Y | 5 | 7 | 9 | 11 | 13 |
- 6) Define Normal and Standard normal distributions. Let X be a normal variate with mean = 30 and S.D. = 5. Find  $P(26 < X < 40)$ . Given the value from standard normal tables:  $P(-0.8 < Z < 2) = 0.765$ . CO3(7)
- 7) Find the lines of regression of Y on X and X on Y for the following data: CO2(14)  

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| X | 1 | 2 | 3 | 4 | 5 |
| Y | 2 | 4 | 5 | 4 | 6 |

OR

For the discrete random variable X with the given probability mass function as below, find value of k, find Mean of X, find Variance of X and find  $P(X=4)$ . CO3(14)

$$f(x) = \begin{cases} k(x^3 + 1); & x = 1, 2, 3, 4, \\ 0, & \text{otherwise.} \end{cases}$$